

BUS 577 – Platform Power – Summer 2026

Session 2: Network Effects: The Engine of Platform Growth

Thursday, July 16, 2026

6:00 – 7:30 PM PT - Zoom

Alex Cheng

Individual Quiz #1

OPENS JULY 21 – DUE JULY 28



Opens July 21, 2026, 12:00 AM PT

Due July 28, 2026, 11:59 PM PT



15-minute time limit

Once started, the clock doesn't stop



2 attempts allowed

Highest of the two scores is kept



Covers Part 1 concepts

Sessions 1–3 material

QUIZ DETAILS

Canvas Quiz

Opens: July 21, 2026 – 12:00 AM PT

Due: July 28, 2026 – 11:59 PM PT

Time limit: 15 minutes

Attempts: 2 – highest score kept

Coverage: Part 1 (Sessions 1–3)

Group Formation – Quick Update

SELF-FORMED GROUPS OF 4-5



Thanks for the emails so far

I've received a number of notes about self-forming your groups of 4-5



Still have a question?

Email BOTH Reza and me – not just one of us – so nothing falls through the cracks

reza_shams@sfu.ca + sze_hang_alex_cheng@sfu.ca

Want to Talk 1:1?

OFFICE HOURS & OUTREACH



Book a 30-minute Zoom slot

A few of you have asked for 1:1 time on the course or your startup idea — happy to do it



Or text / LinkedIn / WhatsApp me

672-971-9091

<https://calendly.com/alexcheng-zenithstudio/30min>

A Note on Missing Class

ATTENDANCE & COMMUNICATION



Please email BOTH Reza and me

I've gotten some Canvas messages and emails about missing classes — for anything class-related, please email BOTH Reza and me



Don't stress about it

Missing a class here or there is completely fine



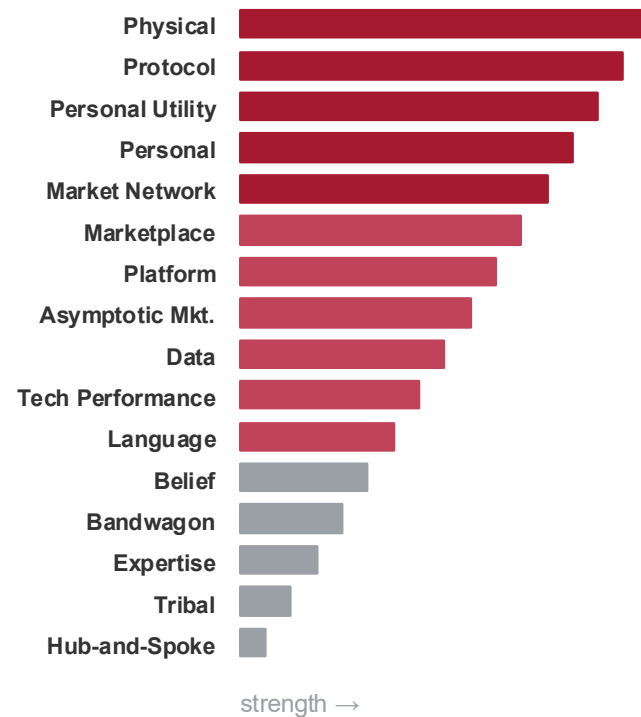
Just two things

Catch up on the readings, and show up ready for the sessions you do attend

The Network Effects Bible

1. THE VOCABULARY

- **Same-side**: more users on your side – can help (Windows) or hurt (Uber surge)
- **Cross-side**: more users on the other side helps you (e.g. Uber: more drivers benefits riders, and back)
- **Indirect**: a bigger catalogue pulls in the other side (eBay, Windows)
- **16 types** total, ranked by strength – physical & protocol are hardest to displace

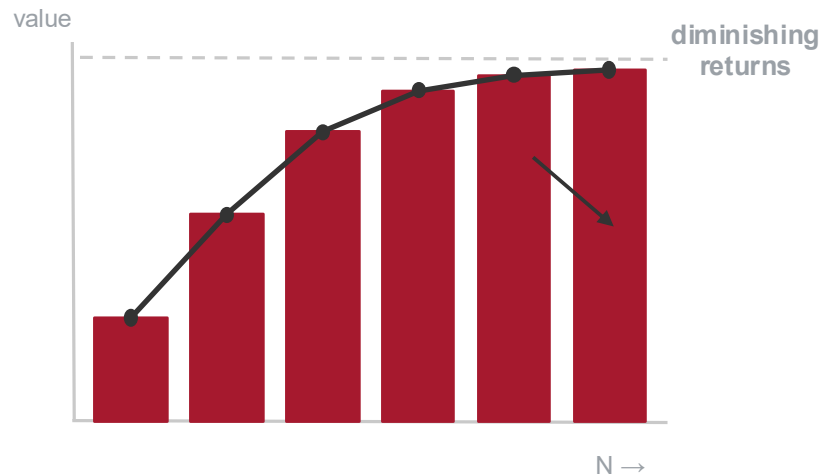


physical & protocol are strongest

The Network Effects Bible

2. GROWTH, STRENGTH, DECAY

- **Metcalf's Law**: value grows with N^2 (communication networks)
- **Reed's Law**: value grows with 2^N – needs clustering / group formation
- **Asymptotic effects**: value can plateau (Uber past a ~4-min wait)
- **Negative effects**: congestion (rush hour) and pollution (cluttered feeds)



e.g. Uber past a ~4-min wait

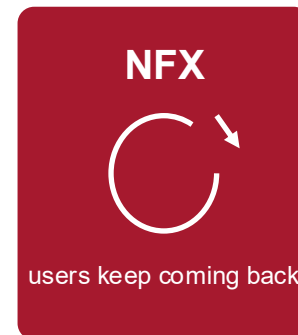
The Network Effects Bible

3. THE TRAP FOUNDERS FALL INTO

- Viral ≠ network effect — a product can have one, both, or neither
- Multiplayer beats single-player — users add value to each other
- Retention, not size, is what makes a network effect real



≠



can have one without the other

Is it a real network effect?

Duolingo

Streaks and leagues feel social —
but you can learn a language alone

**NETWORK EFFECT / NOT A NETWORK
EFFECT**

Zoom

Gets more useful to you as more of
your contacts join

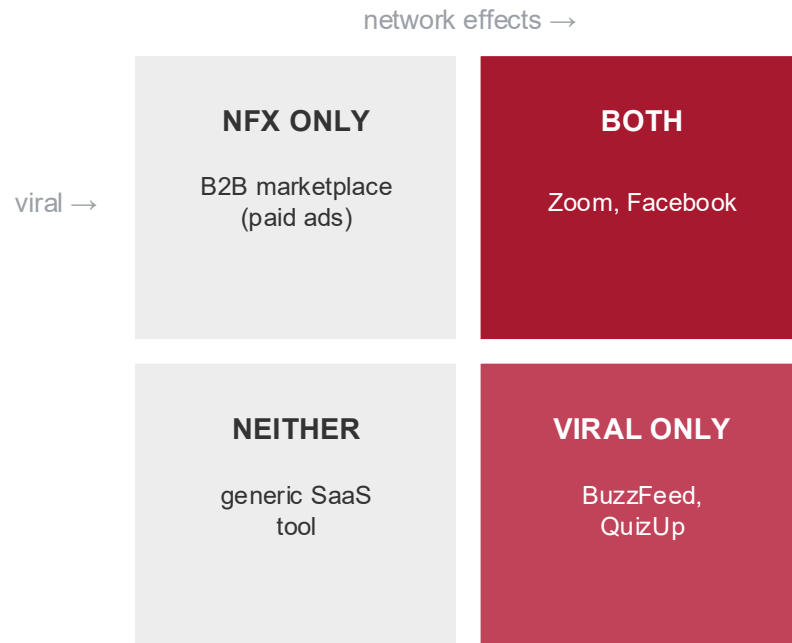
**NETWORK EFFECT / NOT A NETWORK
EFFECT**

Vote now — we'll compare the two.

Viral Effects Are Not Network Effects

1. THE CORE DISTINCTION

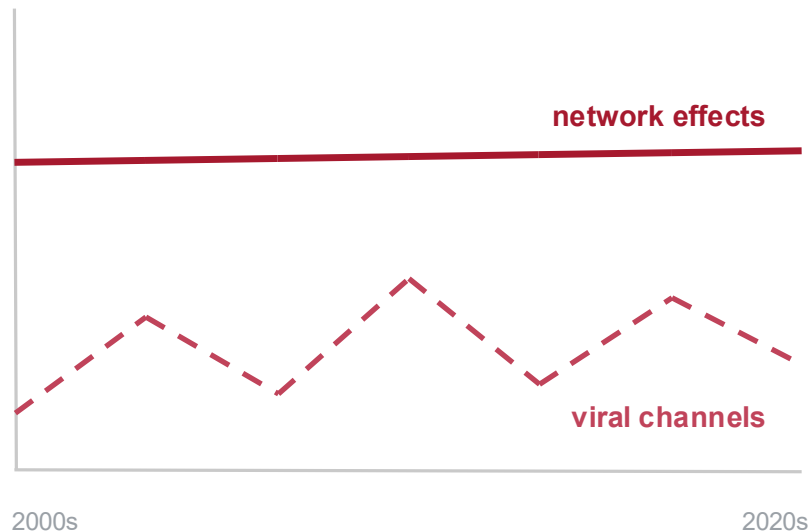
- Viral: existing users bring you new users – a growth mechanism (K-factor)
- NFX: every user adds value, making the product harder to leave – defensibility
- The two can exist independently – Zoom/Facebook (both), B2B marketplace (NFX only), BuzzFeed/QuizUp (viral only)



Viral Effects Are Not Network Effects

2. BUILDING VIRALITY VS. NFX

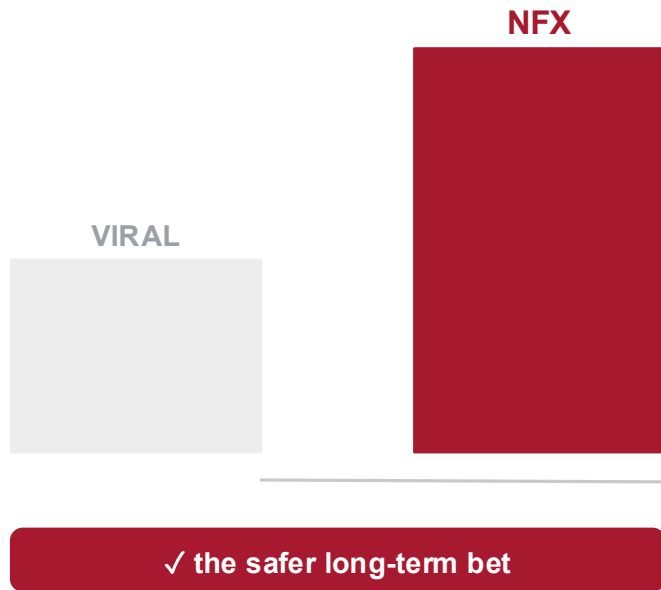
- Virality is built on language – the emotional hook that drives sharing
- Viral channels shift with the times; network effects stay stable for decades
- NFX is math: users constantly weigh “what do I give for what I get”



Viral Effects Are Not Network Effects

3. CONFUSION, AND WHO WINS LONG-TERM

- High-profile examples (Facebook, TikTok) had both – the exception, not the rule
- They can reinforce each other, but that's not the same as being identical
- Viral-only products fade fast (JibJab, BuzzFeed, QuizUp)
- “If you have to choose, choose network effects” – NFX's own conclusion



Viral, network effect, or both?

LinkedIn

Grew via “invite your contacts” —
and gets better as more of your
industry joins

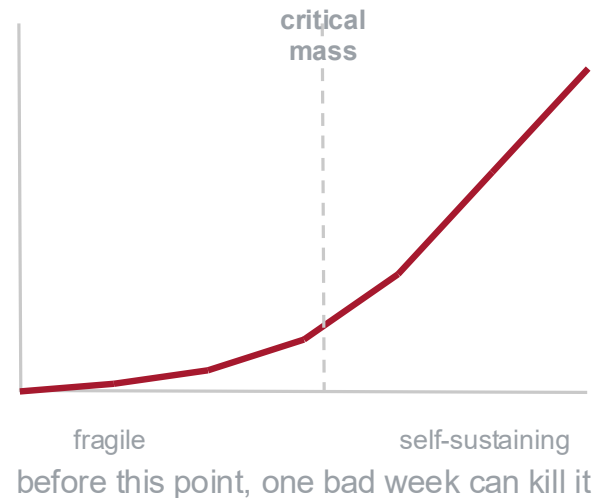
VIRAL / NETWORK EFFECT / BOTH

Vote now — defend your pick.

The Dot-Com Trap & Critical Mass

WHY THIS MATTERS

- NFX, scale, brand, and embedding all get stronger with users – but only NFX makes the product itself better
- Critical mass: the tipping point where network value passes standalone value
- Multi-platform markets survive when multi-homing is easy (Uber + Lyft)



Can This Idea Build Network Effects?

EVALUATING A STARTUP

- Which of the 16 NFX types applies – or does none apply?
- Is this actually scale or brand wearing an NFX costume?
- What's the first thing that breaks as it grows?

☐

Which of the 16 types applies?

☐

Is it really scale or brand in disguise?

☐

What breaks first as it grows?

the same 3 questions, every time

Platform & Network Effect Audit

1. Pick your own business or a startup idea you have in mind
2. Map it: is it closer to a pipeline or a platform?
3. If it's a platform (even partly), who are the producers, consumers, providers, and owners?
4. Map its network effects – same-side, cross-side, indirect – and score each one's strength
5. Identify the single fragility point – where would a competitor attack first?

Raise your hand and share right now – no need to wait or prepare

What we did today

- Network effects vocabulary — same-side, cross-side, indirect — and how to rank their strength
- Viral effects are not network effects — growth vs. defensibility
- The dot-com trap, critical mass, and how to evaluate whether a new idea can build NFX

Next Session

Session 3 – Platform Architecture & Ecosystem Design

How to design the core interaction (pull, facilitate, match), set openness decisions, plug into existing platforms, and avoid depending on one you don't control

Questions or want to talk further? Find me on LinkedIn, send me email or set up a 1:1 Zoom